



**An Assignment on Extracting Various Artifacts of Open Source
Software Systems**

Course Name: Course Name: Software Development Project
Course No: CSE-3106

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Galaxy:

Artifact Name	What is it?	Why it is used?	How is it created?
Github Repository	It's a digital storage space where all the project files, including source code, documentation, and more, are stored and managed.	It serves as a centralized hub for collaboration, version control, issue tracking, and release management.	I can create a GitHub repository by initializing a new repository on the GitHub platform or by pushing an existing project from my local machine.
Code Executable of the Software	It's the compiled or interpreted version of the source code that can be run to execute the software.	Users run the executable to interact with the software and benefit from its functionalities.	It's typically generated through a build process, converting source code into machine-readable instructions or an executable file.
Codebase	The entire set of source code files that make up the software project.	It's the foundation of the software, and developers work with it to add features, fix bugs, and improve the	Developers write and organize code files in a structured manner, following programming languages and

		overall code quality.	project conventions.
Implemented Software Code	The actual lines of code that define the functionalities and behavior of the software.	It's the core of the software, providing the algorithms and logic necessary for its operation.	Programmers write the code using programming languages, adhering to design patterns and best practices.
Programmers	Individuals who write, maintain, and improve the software code.	Programmers are essential for developing and evolving the software, bringing expertise in coding and problem-solving.	Programmers gain knowledge through education and experience, and they contribute to projects by writing code.
Tools	Software applications or utilities used to aid in the development, testing, and maintenance of the project.	Tools automate tasks, ensure code quality, and streamline the development process.	Tools are developed by other programmers or organizations and integrated into the project's workflow

Issues	Items representing tasks, enhancements or other topics that need attention.	Issues help manage work, facilitating communication among team members.	Team members can create issues to discuss any aspect of the project.
Pull Requests	Proposed changes to the codebase submitted by a developer for review and integration.	Pull requests enable collaboration and code review, ensuring code quality before merging changes.	Developers fork the repository, make changes in a branch, and then create a pull request to propose those changes for integration.
Actions	Automated workflows that run based on specified triggers, such as code pushes or pull requests.	Actions automate tasks like testing, building, and deployment, enhancing the development and release processes.	Developers define workflows in a configuration file within the repository, specifying the steps to be executed.
Projects	Boards that help organize and prioritize work items, including issues and pull requests.	Projects provide a visual overview of tasks, making it easier to plan, track progress, and coordinate team efforts.	Users can create and customize projects within the GitHub repository, adding columns and cards to

			represent different work stages.
Security:	Features and tools related to identifying and addressing security vulnerabilities in the project.	Security features help protect the project and its users from potential threats and vulnerabilities.	Developers can configure security settings, use automated tools for vulnerability scanning, and adopt secure coding practices.
Insights	Analytics and visualizations that provide information about the project's activity, contributions, and performance.	Insights help teams understand the project's health, identify trends, and make informed decisions.	GitHub generates insights based on data collected from the repository, including commits, pull requests, and issues.
README	A text file containing essential information about the project, including its purpose, installation instructions, and usage.	README files help users and contributors understand the project quickly and get started with it.	Developers write and update README files to provide relevant and up-to-date information about the project.

Elastic Search:

Artifact Name	What is it?	Why it is used?	How is it created?
Elastic Search GitHub Repository	The GitHub repository for the Elastic search project.	It serves as the central hub for the Elastic search source code, documentation, issues, and collaboration among developers.	Created by initializing a new repository on GitHub and pushing the Elastic search source code to it.
Code Executable (Elastic Search Binary)	The executable binary file for Elastic search.	It allows users to run and deploy Elastic search instances for search and analytics purposes.	Compiled from the Elastic search source code using build tools like Gradle or Maven.
Forks	Copies of the Elastic search repository created by contributors for their own development purposes.	Enables contributors to experiment with changes without affecting the main project until changes are ready for review.	Users click the "Fork" button on the GitHub repository's page, creating a personal copy linked to their account.

Codebase	The entire set of source code files and directories that make up Elastic search.	It provides the foundation for building, extending, and understanding Elastic search functionalities.	Developed by Elastic search contributors using programming languages like Java.
Author	Information about the original author or creator of a specific piece of code or project.	Provides attribution and context about the origin of code, especially in collaborative projects.	Typically, automatically captured by version control systems like Git, and contributors are credited based on their commits.
Labels	Customized labels associated with issues, pull requests, or other GitHub objects.	Offers a flexible way to categorize and prioritize tasks beyond the default labels provided by GitHub.	Defined by maintainers or contributors within the repository settings.
Pulse	A visual representation of the frequency and volume of contributions over time.	Provides a historical view of project activity and highlights periods of increased or decreased contributions.	Automatically generated by GitHub based on historical repository activity.

Activity	A record of recent activity, including commits, pull requests, and issues, visible on the GitHub repository's main page.	To quickly gauge the recent development and engagement within the Elastic search project.	Automatically generated by GitHub based on recent repository actions.
Tools	Software tools and utilities used in the development, testing, and deployment of Elastic search.	Tools assist in various development stages, testing procedures, and automation tasks.	Chosen based on compatibility, efficiency, and best practices in software development.
Pull Requests	Proposed changes to the Elastic search codebase submitted by contributors.	It allows contributors to suggest modifications, improvements, or new features and facilitates code review.	Contributors fork the repository, make changes in a branch, and submit a pull request for review.
Actions	GitHub Actions workflows that automate processes such as testing, building, and deployment.	It ensures consistent and reliable execution of tasks throughout the development lifecycle.	Defined in YAML files within the repository, specifying triggers, jobs, and steps.

